Assignment P1

Julia Nething
nething@gatech.edu

1 QUESTION 1

For the context of this question, I will be reviewing the Georgia Tech registration tool, OSCAR.

1.1 Processor Model Perspective

In this section, I will explain how an expert user and a novice user would register for a class. For either user, first, you must go to buzzport.gatech.edu. Then you click on "Registration (OSCAR)", then "Registration".

If you are an expert user and already have your CRN (Course Reference Number) written down from previous knowledge, then you can click on "Add or Drop Classes". From there, you can input the CRN in the input and click "Submit". So as an expert user, this entire process involves navigating to the website, clicking four buttons, and inputting the CRN for the class that you want.

If you are a novice user and do not know your CRN, this is different process. Back at the Registration page you would instead click on "Look Up Classes", select the term, and then click "Submit". At this point, it is wise to click on "Advanced Search" because the regular course search is not specific enough. On the Advanced Search page, you then click on the relevant subjects, filter by campus (or any other filters you wish), and then click "Section Search". From here, you can either see the CRNs and write them down for future reference (or to go back to the "Add or Drop Classes" page), or you can click the checkboxes next to the classes and click "Submit" to get back to the "Add or Drop Classes" page. Usually when you get back to the "Add or Drop Classes" page you do have to confirm your selections and click "Submit" again, and then you are done registering.

Therefore, from the perspective of the processor model, an expert user can complete the tasks in four clicks plus filling out one input field. A novice user would need to click approximately twelve times (including selecting from the drop-down fields).

1.2 Predictor Model Perspective

From the perspective of the predictor model of the user, the user would predict that clicking on "Registration (Oscar)" the first time would take them to the correct registration page, and that they would not need to click "Registration" again on the next page.

In addition, the user might predict that choosing "Add or Drop Classes" would take them to a page where they could select from a list of classes to either add or drop; instead, it shows them the classes they are already registered for and it shows inputs for the user to input CRNs. If the user did not have prior knowledge of what a CRN was or how to find it, it is unclear from this page how to access the CRN for the class they wish to register.

On the "Look Up Classes" page, the user might predict that this would show them only classes that they were eligible to register for. For example, as an OMSCS student, I thought that that page would show me all classes in OMSCS, not all classes at Georgia Tech. Even after filtering by Computer Science and the Online campus, the list of classes still shows classes that I am not eligible to take (for example, the OMSA section of the class). If a user then attempts to register for one of these classes that is on the list but they are not eligible for, the site does show an error but not a very detailed error. Therefore, they might not understand what they did wrong or understand what action to take next.

1.3 Comparison of Perspectives

The processor model would be focused on cutting down on the number of clicks necessary for registration, so it might suggest consolidating as many sections of the site as possible. For expert users, it could suggest only requiring clicking "Registration" once during the process. For more novice users, it might suggest skipping the term selection page and defaulting to the next term available (perhaps allowing for users to change the term on a separate page).

The predictor model would suggest having the "Add or Drop Classes" page show a list of classes that the user was eligible to register for, and then the user could register for them on the sage page by checking the boxes next to each class. This model also might recommend that the "Look Up Classes" have the correct filters for the logged in student be selected by default.

Therefore, the processor model would be more concerned with limiting the number of clicks or the amount of time needed to register, while the predictor model would be more concerned with showing the user the information they are expecting to see on each page. However, I think both models would consolidate the pages so the user only has to click on "Registration" once.

2 QUESTION 2

2.1 Activity in Multiple Contexts

One activity that I perform in multiple contexts is using the YouTube video app on my phone. Sometimes I use the app to watch a specific video with my full attention, for example to watch a makeup tutorial while sitting on the couch. Sometimes I watch a video while doing another activity. An example of this is when I prop up my phone on the counter to watch a comedy video while I do laundry. In this case, sometimes I glance at the video but I am not watching the video the entire time (more listening than watching). Finally, sometimes I use the app to listen to sleep videos (such as ASMR) to fall asleep. In this case, I am not watching the video at all (I usually put the phone screen-down) but I am listening to it with the intention that I will fall asleep by the end of the video.

2.2 Constraints and Challenges

When I use the YouTube app with my full attention, sometimes I am still distracted (cognitive resources are divided) because I am looking at other aspects of the app such as reading comments on the video, reading the description box, browsing recommended videos, etc. However, most of the time in this context the video has my full attention.

When I use the YouTube app while doing another activity such as laundry, my cognitive resources are divided between the laundry and watching/listening to the video. Sometimes I realize that I missed the beginning of a joke or I missed a physical gesture in the video and have to rewind to get the full context. I often miss the visual aspects of the video because I am instead looking at the laundry, dishes, or some other task.

When I use the YouTube app to aid in falling asleep, I miss the entire visual aspect. This is a unique context because my goal is to not watch the video and to hopefully be asleep by the end of the video, meaning that most of the video will also not be listened to. Additionally, that means that when the video ends I will be asleep, so I do not want the app to auto-play another video or a commercial. If I do need to alter something in the app (for example, if it becomes too loud and I need to turn down the volume), sometimes I am half asleep and unable to find the buttons as quickly or easily.

2.3 Potential Alterations

For the context where I am watching videos with my full attention, the app could be altered to blur text (comments, descriptions, recommended video titles, etc.) until the conclusion of the video. This would help me be able to focus on the video and not get distracted by other text.

In the context where I am watching a video while doing chores, the app could have a feature where it rewinds 10 seconds whenever I say "rewind". Currently there is a button (double tap) that allows the user to rewind by 10 seconds, but often in this context my hands are not free because I am doing a chore, so it would be easier if I was able to verbally give this command.

In the context where I am listening to a video for sleeping, the app could implement a few new features. First, it could disable any commercials that are in the video so that they do not interrupt the sleep sounds. Also, it could remove the video completely (only broadcast the audio) in order to remove any light coming from the phone and to save battery life on the phone. Additionally, if it was somehow possible to know whether the user had fallen asleep or not, it could stop the video once the user had successfully fallen asleep. If it was possible to know that the user was still not asleep at the end of the video, the app could auto-play another sleep video when the first video was done.

3 QUESTION 3

For the context of this question, I will assume that the student has already navigated to the Home page in Canvas for the class that they are taking.

3.1 Gulf of Execution

3.1.1 *Identify Intentions*

The student's overall goal is to upload their assignment to the Canvas portal. Their first sub-goal is to find the assignment page, then they want to upload the assignment, then they want to submit the assignment.

3.1.2 *Identify Actions*

For the first sub-goal, the student will need to click on the Assignments page. Then they need to click on the particular assignment they want to submit. Then they need to click on the file upload and select the correct file. Then they need to click the submit button.

3.1.3 *Execute in Interface*

The student will find the Assignments button in the navigation on the left-hand side or at the top of the page. Then they will find the assignment they want to submit in the list. Up until now, Canvas does a good job of making these buttons clear. However, on the actual Assignment page the student needs to click on "Start Assignment" which seems odd (the user might think that this button would take them to the assignment description instead of the submission). This is the only button available on the page though, which leaves the user no choice but to press it. After they click that button, an upload section is revealed. They can click on "Choose File" to select a file from their computer (there are a variety of other upload options as well). There is a required checkbox on the form (an alert will pop up if the user does not click on this checkbox). However, it is not clear that the checkbox is required because it does not say that it is required and there are no other markings (red text, for example). Once they check that box and click "Submit", they have completed the task.

3.2 Gulf of Evaluation

3.2.1 *Interface Output*

There is no auditory or haptic feedback throughout this process, only visual feedback such as alerts and text.

3.2.2 *Interpretation*

If the user is on the Assignment page and clicks "Submit" without uploading a file or checking the EULA (End-User License Agreement), an alert will pop up that they must "agree to the submission pledge". This is helpful for students to know that that submission was unsuccessful. However, on the page it is called the End-User License Agreement not the submission pledge, which might make it hard for the student to find if they search the page for "submission pledge".

The EULA checkbox has no other indication that it is required (red text, etc.), which also might make it difficult for the user to interpret that that is the checkbox that is required for their submission.

If the user tries to submit without uploading a file, a bright red banner pops up telling them to attach at least one file. This error is easy for the user to understand and fix. Similarly, if the user tries to upload a file type that is not supported, a detailed error shows in red text shows underneath the selector telling them to upload a PDF file instead.

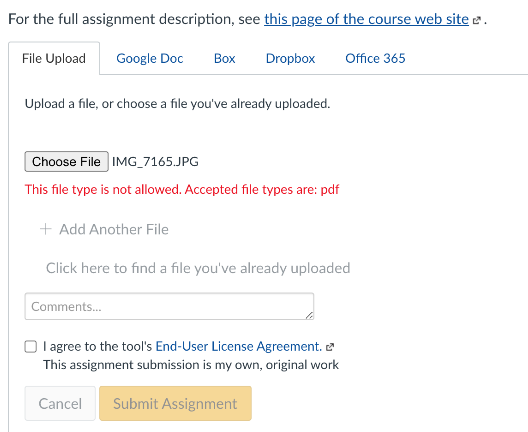

Figure 1—File type error

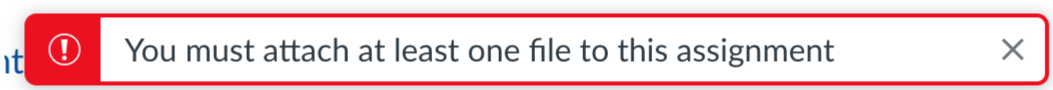

Figure 2—Missing file error banner

3.2.3 *Evaluation*

If the file is successfully uploaded, the page shows a checkmark and "Submitted!" on the right-hand side of the page along with a download link for the file that was submitted. This may not be immediately clear to the user (they might not notice it on the right-hand side of the page), but once they do see it it is clear that the submission was successful. They can also download the file that was uploaded to double-check that it was the correct file.

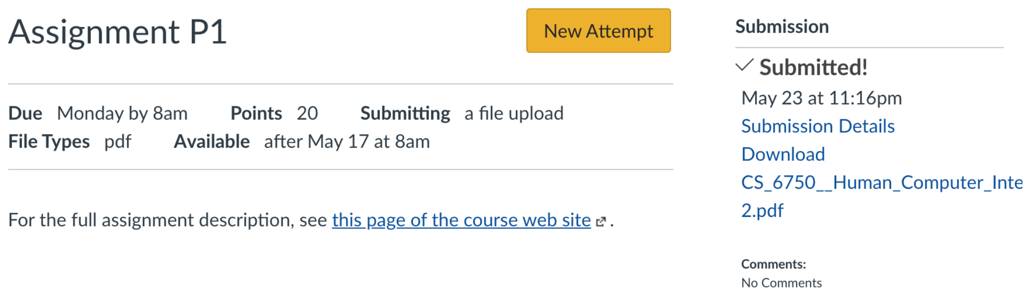

Figure 3—Successful submission

4 QUESTION 4

4.1 Activity with Large Gulf of Execution/Evaluation

As a teacher, I have an appointment calendar on Google Calendar that allows students to book office hour appointment times. Sometimes, for reasons that are not completely clear to me, multiple students are able to book the same appointment time even though they are supposed to be one-on-one sessions. My current theory is that if I try to edit the appointment time block after someone has already booked one of the slots in that block, then it creates a new bookable slot for that same time. So for example, if I have 15-minute slots from 1 PM to 2 PM, and then John Doe books the 1 - 1:15 slot, and then I extend my office hours so that they are now from 1 - 2:30 PM, suddenly there is a new empty slot at 1 - 1:15 even though John's appointment has not been canceled. This has caused a lot of confusion whenever these double-bookings occur, and I often have to email the second student to ask them to book a new time. This is a large gulf of evaluation for me because when I edit the appointment block there is no indication that I have created multiple appointments at the same time. It is not until a student books the second slot that they both appear on my calendar. The only way I would be able to know this is if I constantly compared my personal calendar to my open appointment calendar to try to find overlapping spots. The current interface should help bridge the gap by either not creating the second appointment time at all or by alerting me that someone has already booked that time on my calendar and a new appointment spot will be created if I continue with editing the appointment block. Alternatively, my personal calendar could also show the appointments available side-by-side so I could easily see the overlap as soon as it occurs.

4.2 Activity with Smaller Gulf of Execution/Evaluation

I often use Google Maps for navigation whenever I drive somewhere. If I am going somewhere and will arrive at my destination after it has already closed or within one hour of it closing, Google Maps will alert me when I start the navigation of the destination's closing time. This narrows the gulf because I don't get into a situation where I have arrived at my destination and it is either closing soon or already closed; I can avoid mistakes before they happen. It also helps me understand where I went wrong, because it can show me the hours of operation for every day of the week (in case I didn't realize the business is closed on Mondays, for example) and it can show me other locations' hours of operation (in case I didn't realize this location closes earlier than another location, for example).

4.3 Lessons

The lesson that Google Calendar could learn from Google Maps is to avoid mistakes before they happen. In both situations, I am making a mistake as a user. However, in the Google Maps example the app alerts me of this potential mistake at the beginning of the process so I can be sure that that is the action I want to take. It also explains the reasoning behind the mistake so I can understand where I went wrong (it can show other locations' hours of operation, for example). In the Google Calendar example, I don't find out about the mistake until it's too late, potentially days later and most often not until both students show up at the same office hour. In the Google Calendar example, I am still not even 100% sure what is causing the mistake, I just have a theory, which also makes the gulf wide because I'm not positive how to prevent the problem in the future. In the Google Maps example, I know exactly what the problem is at the beginning. If Google Calendar was able to alert me of my mistakes from the beginning, it would be a much smoother user experience.